

23 — HFO on Both Engines

Proves · HFO on both engines — one fuel factor, applied to both sides.

Mechanics this scenario turns on

- ME and AE each use **their own fuel's** CO2 factor (MGO/MDO 3.93267 · HFO 3.114 · LNG 2.753 · Ammonia 0.35154). The two per-engine cards must sum to the panel total.

Panels described below · What it guards · Verified figures

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 13.

Scenario 19's plant with both fuels set to **HFO** and the fuel price at its default **\$420/ton**.

What it guards

`FuelCo2Factors` carries five fuels. Before this scenario only MDO/MGO (3.93267) and LNG (2.753) had end-to-end coverage — **HFO (3.114) and Ammonia were only ever constants in a config file.**

The plant is deliberately the no-SG one from 19, so the **aux side burns fuel too**. On the original Excel plant the shaft generator covers the whole hotel load and `baselineAE` is zero, which means the aux fuel factor is never applied and a regression there would go unseen.

Verified figures

baseline FOC	8 861.6 t/yr	=	ME 6 880.9	+	AE 1 980.7
baseline CO2	27 595 t/yr	=	ME 21 427	+	AE 6 168

Hand-check

ME	:	6 880.9 × 3.114	=	21 427	✓
AE	:	1 980.7 × 3.114	=	6 168	✓
sum			=	27 595	✓ equals baselineCO2

Both engines use the same factor here, so the sum also equals `totalFOC × 3.114` — which is exactly the legacy single-constant behaviour. That is the point: **HFO on both sides must collapse to the old arithmetic**, and 24 then proves it does *not* collapse when the fuels differ.

Takeaway: the cheapest fuel per ton is not the cleanest per ton — HFO is 21 % below MDO on CO2 while costing 46 % less. The model reports both independently.

